SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS: **Total Marks: 20**

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

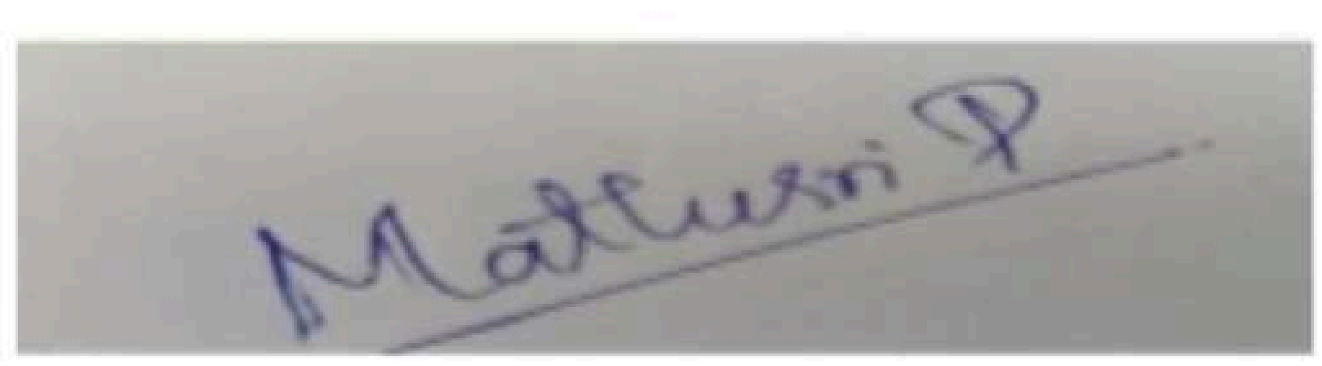
Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I Mathusri P-192311363 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Mathusri P
192311363

CSA4305 - Internet Programming

CO2_AT3_Short Answer Test.

① Responsive web Design and Adaptive layout using css / Flexbox /Grid :-

* Introduction :-

* In modern web development, user access website through that automatically adjust their layout, images, and content according to different screen sizes such as mobile, tablet, and desktop. it provides a better user experience by making websites flexible and easy to access on all devices.

* Importance of Responsive web Design :-

* provides a consistent user experience across different devices
* improves websites accessibility and usability
* Reduces the need for separate mobile and desktop websites
* improves websites performance and search engin ranking

* key css Techniques for responsiveness

① Flexible Layouts.

* Flexible layout use relatives unit like %, em, and rem instead of fixed values. This allows webpages elements to resize automatically.

(2) media Queries : media queries apply different css styles based on device screen size.

Example:

```
@media (max-width : 600px) {
    body {
        font-size : 14px;
    }
}
```

* (3) css Flexbox :- is an one dimentional layout system used to arrange webpage elements in rows or columns. it auto-matically adjust the size and position of elements

Example:

```
.container {
    display : flex;
    flex-wrap: wrap;
}
```

* (4) css Grid :- is used to create layout using rows and columns it is suitable for designing complex pages structure.

Example:

```
.container {
    display : grid;
    grid-template-columns: repeat (3, 1fr);
}
```

* Sample Responsive Layout (HTML)

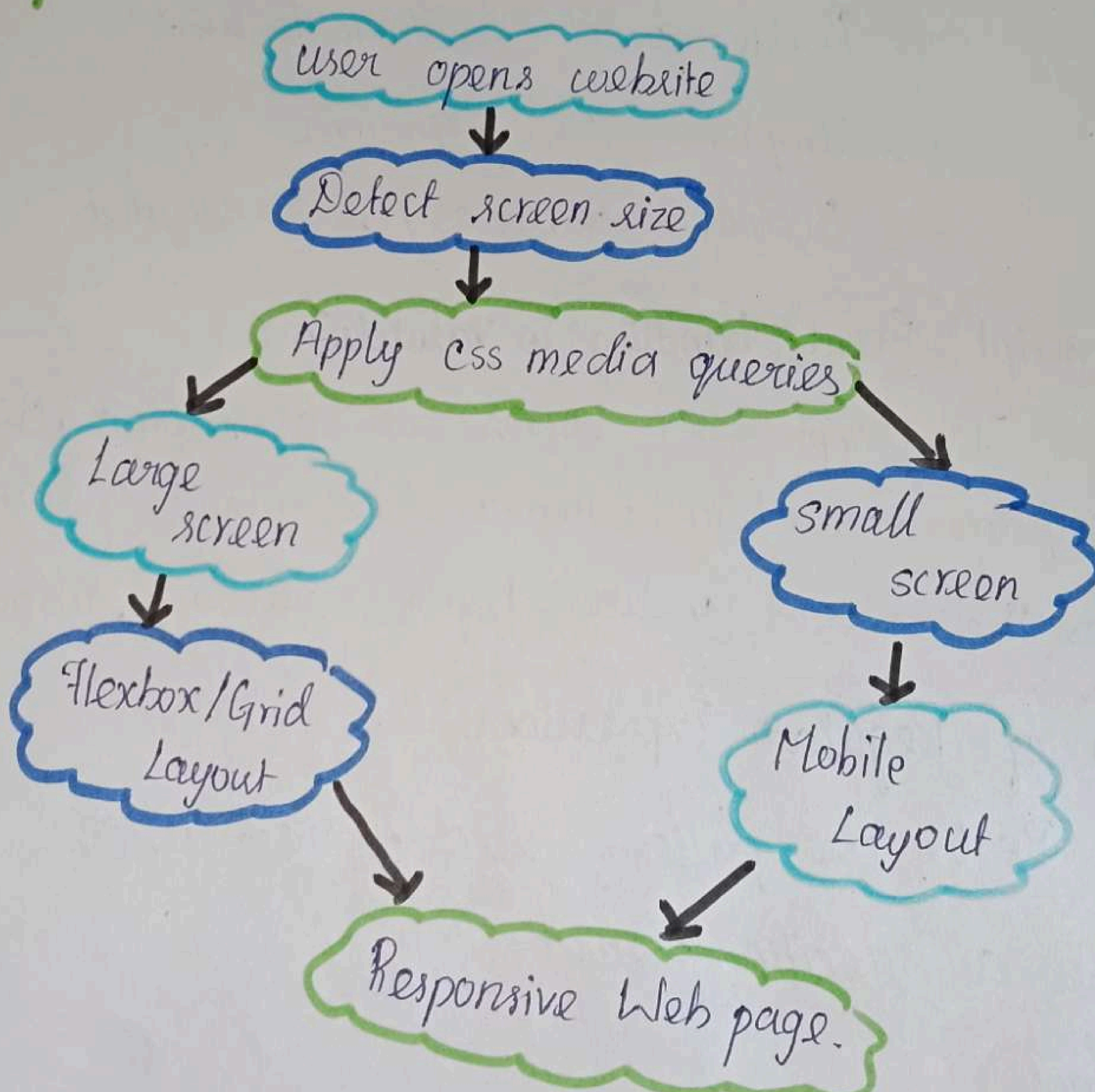
```
< dive class = "box'>
    <div> Header </div>
    <div> Content </div>
    <div> Footer </div>
</div>
```

CSS:

```
.box {
    display : flex ;
    flex - wrap : wrap;
}
```

```
@media (max - width : 600px) {
    .box {
        flex - direction : column;
    }
}.
```

* Flowchart:-



2) Client side validation using javascript and Regular Expression

* concept : client-side validation:-

* client side validation is a process of checking user input data in the browser before sending it to the server. it to the server : it helps to ensure the user letter enter correct and complete information in a form.

* The main purpose of client side request con validation is:

→ Reduce incorrect data entry

→ Provide quick feedback to users

→ Improves user experience

→ Reduce unnessary server request

* Javascript Event handling in Validation

* Javascript user events such as submit, click and input to perform validation. when a user submit a form javascript Checks the entered values before allowing submission

* Roles of Regular Expressions.

* RegEx are pattern used to check whether user input matchs a specific format.

Example:

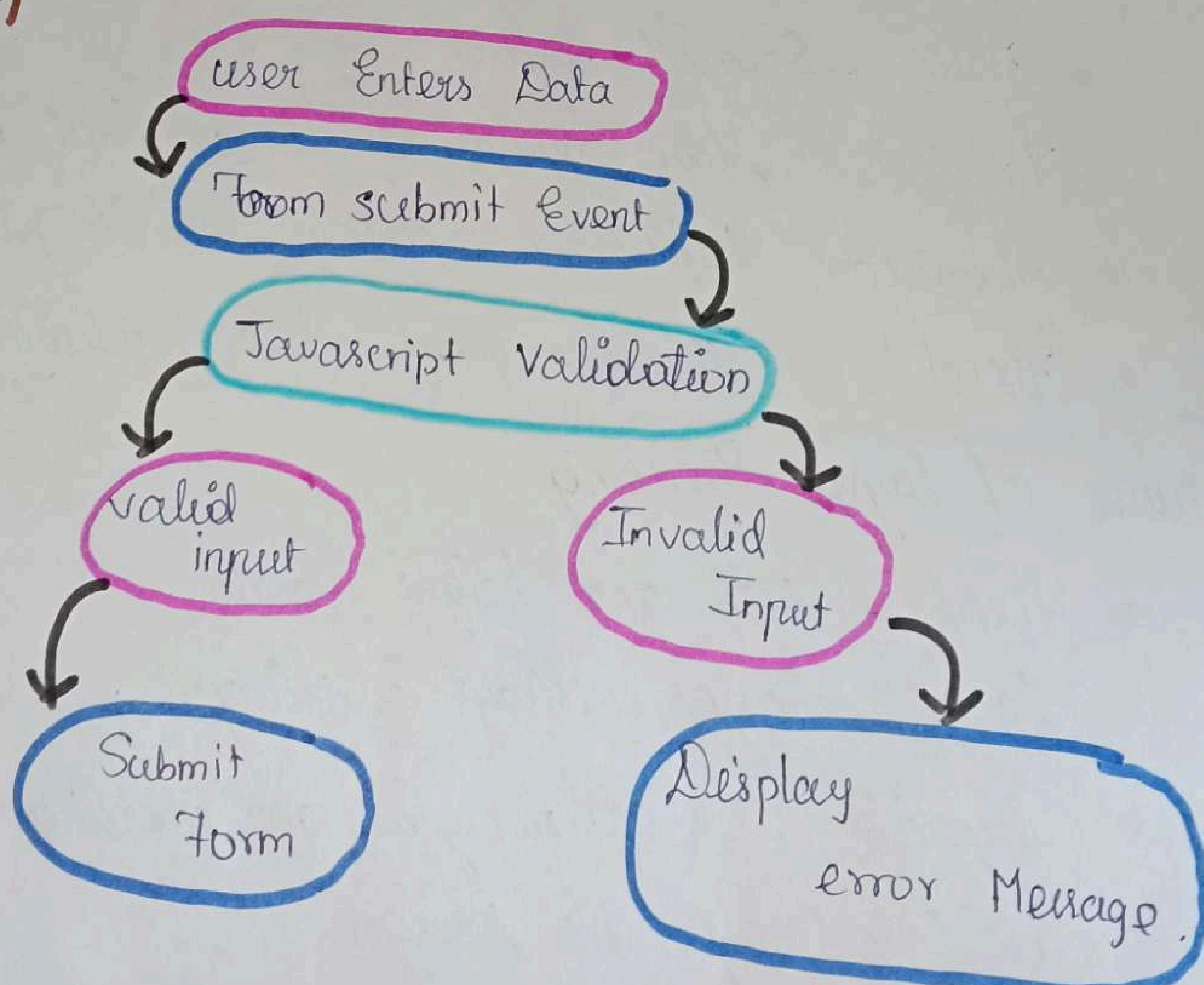
* Email validation : checks username, @symbol, and domain
* Phone Number validation: checks only numbers & correct length.

* Sample JavaScript Validation :-

```
<input id="email">
<button onclick="check()">submit</button>

function check(){
  let email = document .getElementById ("email").value;
  if(email.includes("@")){
    alert("valid Email);
  }
  else{
    alert(" Enter valid email");
  }
}
```

* Flowchart.



③ CSS Box model, Positioning and Media Queries

* Concept : Css Box Model:
  * The CSS Box Model describes how every HTML element is displayed as a rectangular Box . it consists of four component.
    * Content - The actual text (or) image
    * Padding - space between the content and border
    * Border - surround the padding and content
    * Margin - Space outside the border.

* Css Positionings Techniques:
  * Css positioning controls where element appear on a webpage.
    * static - Default position
    * Relative - Moves relative to its original position
    * Absolute - Positioned relatives to its parent element.
    * Fixed - Remains fixed while scrolling.

* Causes Of Layout Breaking
  * Fixed width and height values
  * Large images without resizing
  * improve use of margins and padding
  * miiing responsive Design.

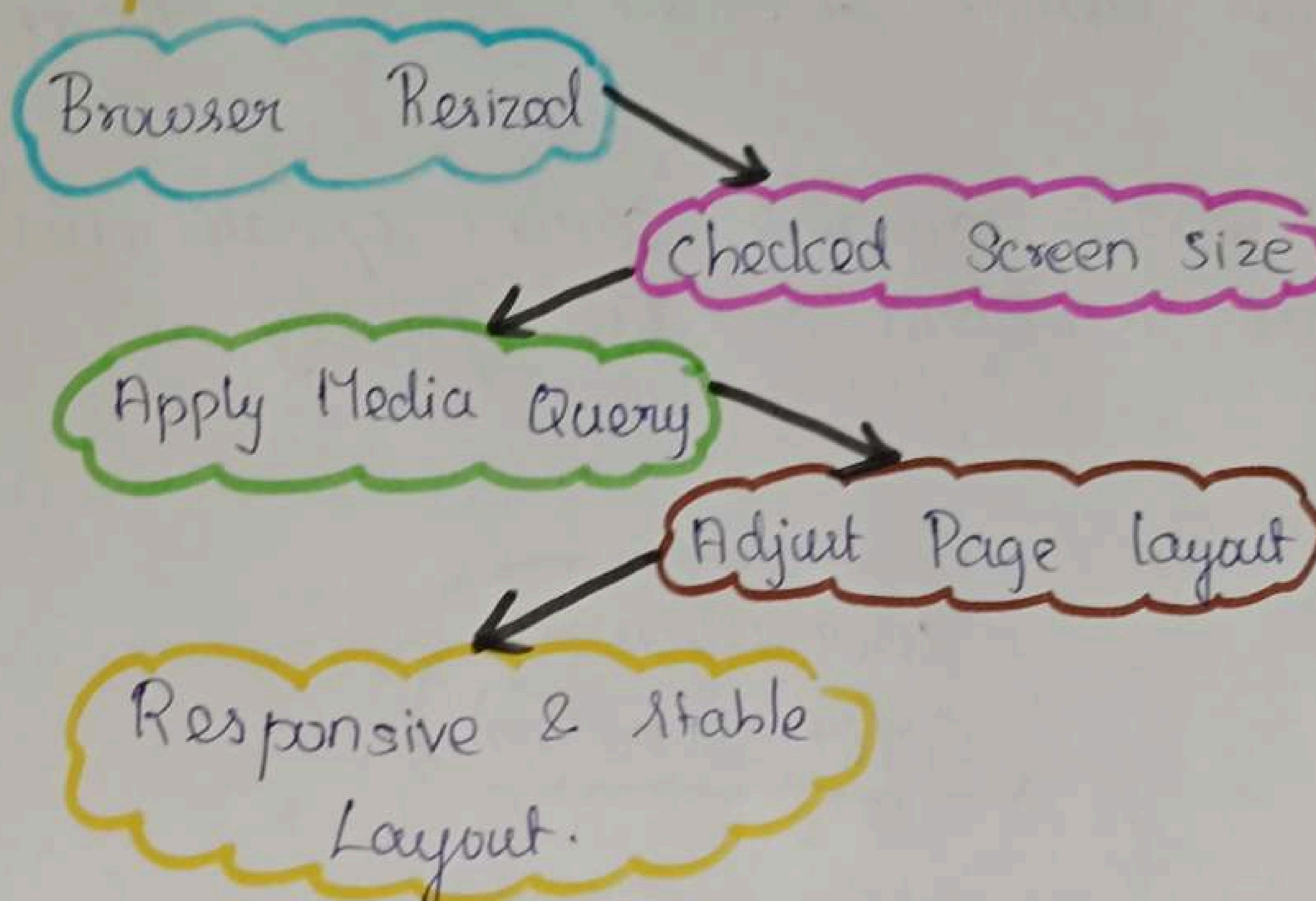
Media Queries.

* Media queries change the website (or) webpage layout based on Screen size.

* **Example:**

```
@media (max-width : 600px) {
    .container {
        width : 100% ;
    }
}
```

* **Flowchart:-**



* **Sample CSS:-**

```
.box {
    width : 300px ;
    padding : 20px;
    border : 2px solid black;
    margin : 10px;
}
```

* **Conclusion:-**

* The CSS Box model and positioning techiniques helps organize webpage layouts. Media queries ensure the layout remain stable and resposive on different Device.

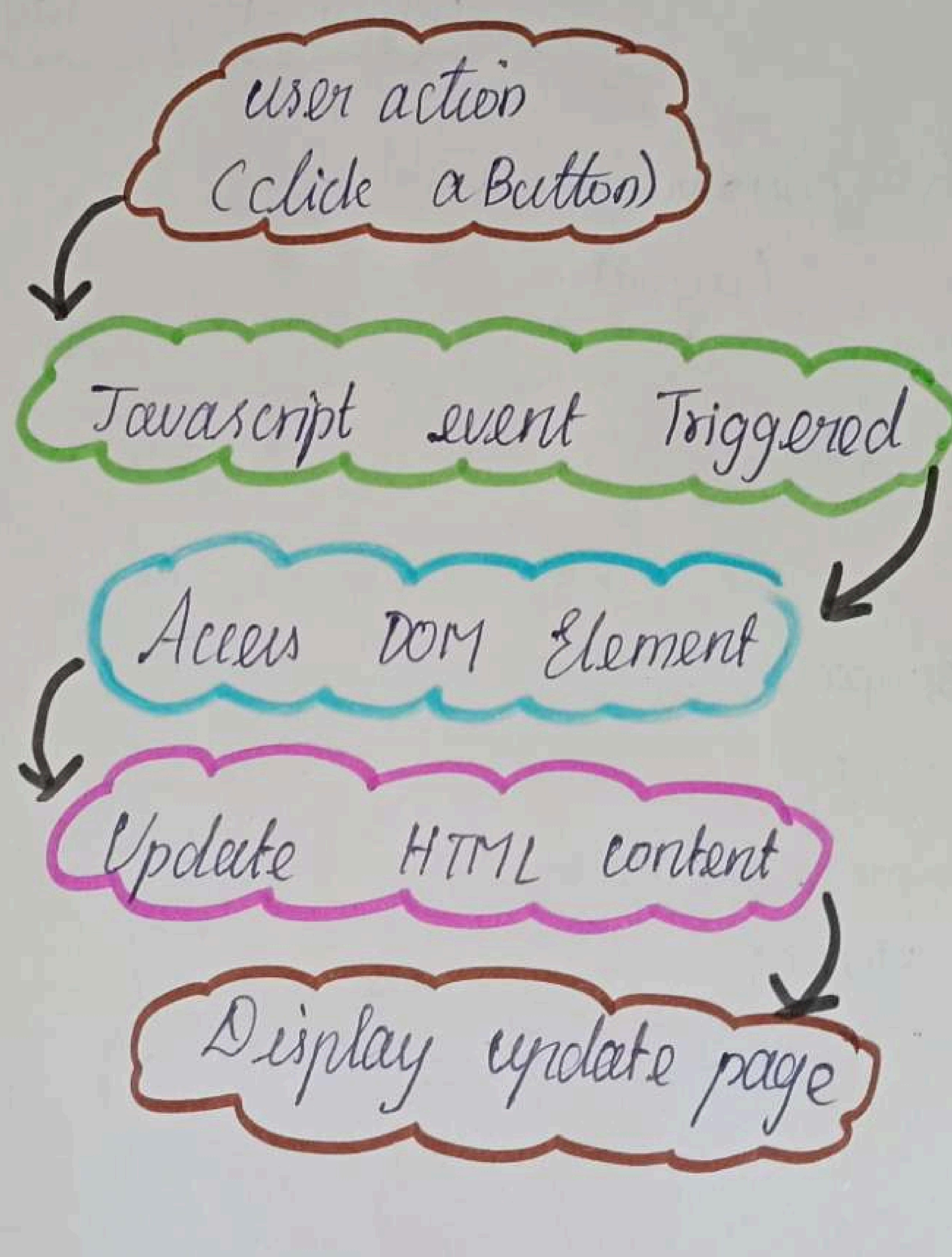
④ Document Object model (DOM) and Dynamic Content update.

* **Concept:-**

* The Document object Model (DOM) is a programming interface for HTML documents. it represents a webpage as a tree structure, where each HTML element is treated as an object (node). Javascript uses the DOM to access, modify add, or remove elements without reloading the entire webpage.

* The DOM enables dynamic content updates, making websites more interactive and user-friendly.

* **Flowchart**



* **Example:**

HTML:

```
<p id = "msg" >welcome </p>
<button onclick = "changeText()" >Click Here</button>
```

Javascript:-

```
function changeText(){
    document.getElementById("msg").innerHTML=
                  "welcome to internet programming";
    }
```

Explanation:-

- Initially, the webpage displays "welcome". when the user clicks the button, javascript accesses the DOM using gentlelementById() and changes the paragraph text to "welcome to interpet programming" without refreing the page.

Improvement:-

- use Efficient DOM method to improve performance
- update only the required elements instead of reloading the entire page
- Validation user input before updating the DOM
- use event listener for better code organization.

Conclusion:-

- The Document Object Model (DOM) allows javascript to dynamically updat webpage content without reloading the page DOM. manipulation improves website interactive, response faster.